

Louen Pottier

AI Research Engineer • Scientific ML, Physics & 3D Vision

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PROFESSIONAL EXPERIENCE

CEA-List, Paris-Saclay

since 2022

AI Research Engineer

Neural network architectures for real-time physical simulation and model-order reduction in virtual reality (physics-informed NNs, surrogate models, world models, Gaussian Splatting).

- **LaGSplat** (patent filed, sole inventor): physics-governed interactive simulation learned from a few seconds of monocular video. Real-time web demo (louenpottier.github.io/lagsplat.html).
- **LEBNN**: interpretable model-order-reduction architecture, deployed in the European programme ARISE to simulate soft-robot components in real time.
- **Industrial project with Renault**: GNN crash-test surrogate from scarce data, computing time cut by several orders of magnitude.
- **Industrial project with Forvia**: physical-field surrogate for design optimisation.

Independent AI R&D

AI Research Engineer (freelance)

- **Where** (2025): GPS-free indoor geolocation under severe noise. Full pipeline designed from scratch; localisation error reduced by 70 % on the hardest industrial cases, building blocks integrated into the commercial product (PyTorch, ONNX).
- **Arkose** (2024): 3D reconstruction of a climbing wall from a single photograph (monocular depth estimation + 3D Gaussian Splatting).
- **SENS Conseil / CReSTIC** (2020–2022): graph + recurrent networks for PLC language processing and formal verification; published in *MDPI Processes* (2023).

EDF Lab Paris-Saclay (EDF R&D) / Centre Borelli

2021

Research Intern

- Hierarchical GNN with continuous-kernel convolutions (SplineConv) predicting vortical flows on irregular meshes. Presented at the NeurIPS 2021 ML & Physical Sciences workshop.

CEA Saclay (LM2S) / Centre Borelli

2020

Research Intern

- LSTM reduced-order model of a nonlinear hysteretic joint coupled to the Cast3M finite-element code: computing time reduced by 95 %.

PUBLICATIONS

Journal articles

- **L. Pottier**, A. Thorin, F. Chinesta. Latent-Energy-Based NNs: an interpretable neural network architecture for model-order reduction of nonlinear statics in solid mechanics. *Journal of the Mechanics and Physics of Solids*, vol. 194, 2025. [DOI]
- E. Siboulet, **L. Pottier**, T. Ranger, B. Riera. Fresh approaches for structured text programmable logic controllers programs verification. *MDPI Processes*, vol. 11(3), 2023. [DOI]

Workshops and posters

- A. K. Agrawal, **L. Pottier**, A. Thorin. Latent-Lagrangian neural networks for model-order reduction of nonlinear dynamical systems. *MORTech 2025*.
- **L. Pottier**, A. Thorin. Designing recurrent cells to enforce stability in real-time dynamical simulations. Poster, *IUTAM Symposium on Data-driven Mechanics*, 2022. [HAL]
- L. Meyer, **L. Pottier**, A. Ribes, B. Raffin. Deep surrogate for direct time fluid dynamics. *NeurIPS 2021, Workshop on Machine Learning and the Physical Sciences*. [arXiv]

Preprints (2026)

- **L. Pottier**. LaGSplat: inferring physics-governed interactive simulation from monocular video using latent Lagrangian Gaussian splatting. [arXiv]
- **L. Pottier**, L. Lesueur, A. Thorin. Partially linear autoencoders for manifold learning and dimensionality reduction. [arXiv]

TEACHING

ESILV, Paris

since 2022

Part-time Lecturer

- Fluid Mechanics (Year 3, 10 groups of 30), Thermodynamics (Year 2, 11 groups of 30), Partial Differential Equations (Year 3, 2 groups of 30), Advanced Fluid Mechanics (MSc Year 1, 2 groups of 30, in English).
- About 750 students taught, in French and in English.

EDUCATION

CentraleSupélec (Université Paris-Saclay)

2021–2022

M.Sc., Mechanical Simulation of Structures & Coupled Systems

ENS Paris-Saclay

2018–2021

- Diploma in Artificial Intelligence (selective programme, 2019–2021): MVA coursework, research year co-supervised by Centre Borelli.
- M.Sc. Year 1, Mechanics of Materials and Structures (2019–2020).
- B.Sc. SAPHIRE, Mathematics, Physics & Engineering Sciences (2018–2019).

Lycée Blaise-Pascal, Rouen

2016–2018

Classe Préparatoire aux Grandes Écoles (intensive two-year maths & physics programme)

AWARDS

- **Mistral Vibe Hackathon** (2026): 1st in category, 3rd overall. *Impostral*, a multiplayer Turing test where Mistral agents infiltrate a group of humans (agentic AI, FastAPI, STT/TTS). impostral.com

SKILLS & LANGUAGES

ML & SciML	PINNs, surrogate models, representation learning, 3D/4D Gaussian Splatting
Model deployment	PyTorch, C++, ONNX, TorchScript, libtorch, Docker
Physics & simulation	Continuum mechanics, model-order reduction, FEniCS, Cast3M, Ansys
Languages	French (native), English C1 Advanced (Cambridge English Qualification)